

Insup Lee

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Research Interests

AI for security, Security for AI, Data augmentation, Generative models.

Education

Korea University Seoul, Korea
Ph.D. in Information Security Sep 2019 - Aug 2026

- Dissertation: Domain-Specific Generative Models for Data Augmentation in Cybersecurity
- Advisors: Sangjin Lee and Seokhie Hong

Korea University Seoul, Korea
B.E. in Cyber Defense Mar 2014 - Feb 2018

Employment History

KAIST Seoul, Korea
Postdoctoral Fellow, LLM Innovation Research Center Sep 2026 - Present

- Advisor: Dongsu Han (INA Research Group)

Korea University Seoul, Korea
Lecturer Sep 2025 - Feb 2026

- Taught graduate-level course “Computer Networks”

Ministry of National Defense Seoul, Korea
Cyber Officer Aug 2023 - May 2025

- Deployed to Abu Dhabi, UAE for joint research on AI-based drone security (Apr 2024 - Mar 2025)

Agency for Defense Development (ADD) Seoul, Korea
Researcher, Cyber Technology Center Jul 2018 - Jul 2023

- Advisor: Changhee Choi

Publications

(C: conference, J: journal, P: preprint †: corresponding author)

Preprints

- P1 *Schema-Agnostic Knowledge Graph Construction via Hybrid Ontology Discovery for Cyber Threat Intelligence*
S. Kim, J. Kim, D. Kang, D. Kim, and **I. Lee**[†]
arXiv:2606.01208, May 2026.

International Publications

- C4 *Mnemonic Hack: Recovering the Master Seed from Bitcoin Hardware Wallets via Side-Channel Attacks*
J. Baek, G. Ahn, S. Park, D. Bae, G. Kim, **I. Lee**, S. Hong, and H. Kim
ACM Conference on Computer and Communications Security (**CCS’26**), Nov. 2026.
- C3 *Exploiting Per-Core Leakage: Electromagnetic Side-Channel Monitoring of Multicore Architectures*
D. Bae, S. Park, **I. Lee**, Y. Jung, K. Lee, H. Kim, and S. Hong
ACM/IEEE Design Automation Conference (**DAC’26**), Jul. 2026.
- J9 *MuCamp2: Generating Validated Cyber Campaign Variants via Constrained LLMs for Group Attribution*
K. Kim, J. Lee, **I. Lee**, I. Han, J. Lee, S. Shim, and C. Choi

IEEE Access, Vol. 14, pp. 87329-87344, Jun. 2026.

J8 *LeakDiT: Diffusion Transformers for Trace-Augmented Side-Channel Analysis*

I. Lee, D. Bae, S. Hong, and S. Lee

IEEE Computer Architecture Letters, Vol. 25, No. 1, pp. 5-8, Jan./Jun. 2026.

J7 *Multi-Step LLM Pipeline for Enhancing TTP Extraction in Cyber Threat Intelligence*

H. Kim, D. Lee, I. Lee, S. Lee, and S. Lee

IEEE Access, Vol. 13, pp. 179696-179710, Oct. 2025.

J6 *Enhancing Modulation Classification via Diffusion Transformers for Drone Video Signal Processing*

I. Lee[†], K. Alteneiji, and M. Alghfeli

IEEE Signal Processing Letters, Vol. 32, pp. 3325-3329, Aug. 2025.

J5 *MuCamp: Generating Cyber Campaign Variants via TTP Synonym Replacement for Group Attribution*

I. Lee and C. Choi

IEEE Transactions on Information Forensics and Security (TIFS), Vol. 20, pp. 6162-6174, Jun. 2025.

J4 *UniQGAN: Towards Improved Modulation Classification With Adversarial Robustness Using Scalable Generator Design*

I. Lee and W. Lee

IEEE Transactions on Dependable and Secure Computing (TDSC), Vol. 21, No. 2, pp. 732-745, Mar./Apr. 2024.

J3 *Camp2Vec: Embedding Cyber Campaign With ATT&CK Framework for Attack Group Analysis*

I. Lee and C. Choi

ICT Express, Vol. 9, No. 6, pp. 1065-1070, Dec. 2023.

J2 *Exploiting TTP Co-occurrence via GloVe-Based Embedding With ATT&CK Framework*

C. Shin, I. Lee, and C. Choi

IEEE Access, Vol. 11, pp. 100823-100831, Sep. 2023.

J1 *BAN: Predicting APT Attack Based on Bayesian Network With MITRE ATT&CK Framework*

Y. Kim, I. Lee, H. Kwon, G. Lee, and J. Yoon

IEEE Access, Vol. 11, pp. 91949-94968, Aug. 2023.

C2 *Anomaly Dataset Augmentation Using Sequence Generative Models*

S. Shin, I. Lee, and C. Choi

IEEE International Conference on Machine Learning and Applications (ICMLA'19), Dec. 2019.

C1 *Opcode Sequence Amplifier Using Sequence Generative Adversarial Networks*

C. Choi, S. Shin, and I. Lee

International Conference on ICT Convergence (ICTC'19), Oct. 2019.

Under Review

- *Preserving Statistical Leakage Structure in Side-Channel Trace Generation via Domain-Specific Diffusion Transformers*
I. Lee, D. Bae, G. Kim, I. Lee, S. Hong, and S. Lee
Submitted.
- *MoDiT: Domain-Specific Diffusion Transformers for Automatic Modulation Classification*
I. Lee[†] and M. Alghfeli
Submitted.
- *Multi-Domain Side-Channel Analysis for Anomaly Detection in Embedded System*
S. Park, D. Bae, I. Lee, J. Kim, H. Oh, H. Kim, and S. Hong
Submitted.

Domestic Publications (Korean)

- D2 H. Park and I. Lee[†], "Enhanced DDoS Detection via Traffic Volume-Based Labeling and Transfer Learning," Journal of Internet Computing and Services (JICS), Vol. 26, No. 4, pp. 1-8, Aug. 2025.
- D1 K. Kim and I. Lee[†], "User Behavior Embedding via TF-IDF-BVC for Web Shell Detection," Journal of The Korea Institute of Information Security & Cryptology (JKIISC), Vol. 34, No. 6, pp. 1231-1238, Dec. 2024.

Patents

International Patents

- C. Choi and **I. Lee**, "Method of Augmenting Cyber Attack Campaign Data to Identify Attack Group, and Security System Performing the Same," U.S. Patent Application Number. 19/244,041.
- C. Choi, C. Shin, S. Shin, S. Seo, and **I. Lee**, "Method for Training Attack Prediction Model and Device Therefor," U.S. Patent Application Number. 18/126,005; U.S. Patent Number. US20230308462A1, September 28, 2023.

Domestic Patents

- C. Choi and **I. Lee**, "Method for Augmenting Cyber Attack Campaign Data to Identify Attack Group, and Security," Korea Patent Application Number. 10-2024-0176082, December 2, 2024.
- C. Choi, **I. Lee**, C. Shin, and S. Lee, "Information Identification Method and Electronic Apparatus Thereof," Korea Patent Application Number. 10-2024-0006106, January 15, 2024.
- C. Choi, S. Shin, and **I. Lee**, "Apparatus, Method, Computer-readable Storage Medium and Computer Program for Generating Operation Code," Korea Patent Application Number. 10-2019-0141865, November 07, 2019; Korea Patent Number. 10-2246797, April 30, 2021.

Other Experience

- AI Cyber Challenge (AIXCC), DARPA and ARPA-H, USA Apr 2024 – Aug 2024
- Participated in the semifinal round as a member of Team KORIA, submitting our cyber reasoning system that leverages LLMs for automated detection and patching of software vulnerabilities
- SW Outsourcing Development, KCMVP-Certified Cryptographic Module Jun 2017 – May 2018
- Implemented a cryptographic module with 25,000 LoC in C - ARIA block cipher (modes: ECB, CBC, CTR), hash functions (SHA-256, SHA-512), and HMAC-based DRBG for Windows (.dll) and Linux (.so)

Awards and Honors

- KU Graduate School Achievement Award Feb 2026
Korea University, Seoul, Republic of Korea
- Certificate of Commendation (UAE-ROK Engagement Program) Mar 2025
United Arab Emirates Ministry of Defense
- Ambassador's Commendation Mar 2025
Embassy of the Republic of Korea to the United Arab Emirates
- Full Tuition Scholarship (Korea University) Mar 2014 – Feb 2018
Ministry of National Defense, Republic of Korea

Professional Activities

Reviewer

- IEEE Transactions on Dependable and Secure Computing (TDSC), 2025
- IEEE Transactions on Information Forensics and Security (TIFS), 2026
- IEEE/ACM Transactions on Networking (ToN), 2026
- IEEE Transaction on Communications (TCOM), 2025, 2026
- IEEE Journal on Selected Areas in Communications (JSAC), 2025, 2026

Teaching Experience

- Lecturer, Computer Networks (SCS302, Fall 2025), Korea University Sep 2025 - Dec 2025

Aug 2026

- Instructor, Penetration Testing - Intermediate, UAE Ministry of Defense Sep 2024
- Instructor, Penetration Testing - Basic, UAE Ministry of Defense Jul 2024

Invited Talks

- Cybersecurity and Hacking in the AI Era March 2026
Major Exploration for Freshman, Sungshin Women’s University
- Exploring Career Paths in Security Research November 2025
Major Exploration for Freshman, Sejong University

Mentoring Experience

- **Seonwoo Kim** (Major at Ministry of National Defense) Feb 2026 – Present
Project: LLM-based knowledge graph construction for CTI [P1]
- **Sujin Park** (Ph.D. Student at Korea University) Jun 2025 – Jul. 2026
Project: Side-channel analysis for anomaly detection
- **Hyoungrok Kim** (M.S. Student at Korea University) Mar 2025 – Sep 2025
Project: LLM-based TTP extraction for CTI [J7]
- **Hyunjun Park** (Navy Lieutenant at Ministry of National Defense) Nov 2024 – Feb 2025
Project: DDoS detection via transfer learning [D2]
- **Kangmun Kim** (First Lieutenant at Cyber Operations Command) Jan 2024 – Sep 2024
Project: Web shell detection via user behavior embedding [D1]